Lesson plan · for the teacher

Computational thinking, unplugged and with blocks: unit check

Programming · Level 1 · Unit check

Time	Materials
45 min	Board, chalk, paper. No devices needed.

Learning goals

- Algorithms: exact instructions
- Loops
- Decisions and debugging
- Break it down

Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Act out the robot questions on a chalk grid

Discussion

- Which game will you play with your class first?
- Which question surprised you, and why?
- Which idea will your pupils find hardest?

Exit ticket

- One unplugged activity I will run, and when

Try it in class tomorrow

Pick one game from this course (the human robot, clap the loop, Simon says IF or guess my number) and play it with a class this week. Then tell a colleague how it went.

Worksheet · for pupils

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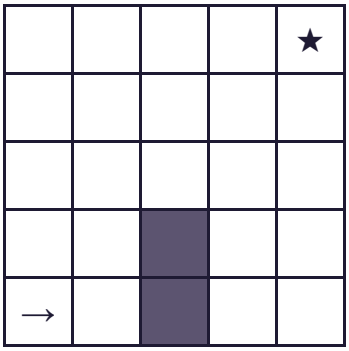
Name _____ Class _____ Date _____

1 **What is an algorithm?**

◯ A computer program that is very fast ◯ Exact steps, in order, that finish and solve a problem

◯ A guess that is usually right ◯ A kind of computer

2 **Walk the robot around the wall to the flag, with 5 cards at most.**



Cut out cards (↑ forward, ↶ turn left, ↷ turn right) and lay out a program that takes the robot to the ★. Write it here:

3 **A loop makes a program longer, but easier to read.**

◯ True ◯ False

4 **Match each part of computational thinking to an example.**

Decomposition	• • • •	A recipe for bread
Pattern recognition	• • • •	A timetable that shows subjects, not what is taught
Abstraction	• • • •	Noticing the canteen queue is longest every Monday
Algorithm	• • • •	Planning sports day as events, teams and timings

5 **The robot starts facing up and runs this program. Which way is it facing at the end?**

```
for i in range(3):
    forward()
    right()
```

◯ Up ◯ Right ◯ Left ◯ Down